

University of the Witwatersrand

Health Analytics for Data Science (COMS5027A) & Health Analytics (FAMH4004A)

Technical Report

Assignment 1: District-Level TB Retention Analysis [WEEK 1 TO 3]

Which variables are most strongly associated with a district's inability to retain drug-susceptible Tuberculosis (TB) patients in continuous care in 2024?

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1. Introduction and data source:

For Tuberculosis(TB) treatment to be effective patients must keep on getting continuous care. Patient retention is very essential for managing TB since treatment can be stopped if patients are **lost to follow up**. Thus, The purpose of this investigation was to determine which variables are the one that are closely related to the district's failure to retain drug- susceptible TB patients in continuous care in 2024.

The dataset used for this project came from the Health Systems Trust's District Health Barometer(DHB) 2024. The extracted dataset includes 11 variables and 52 South African Health districts. **TB DS client lost to follow up rate (LTFU)** is defined as the target variable, **and the greater rate simply indicates a lower retention rate**.

As the possible explanatory variables, 8 of the numerical variables were examined. The variables were:

- TB DS death rate
- TB/HIV co-infected client on ART rate ([ETR.Net](#))
- % of target tested for TB
- Professional nurses per 100 000 population
- Provincial & LG District Health Services expenditure per capita (uninsured)
- Provincial & LG PHC expenditure per PHC headcount
- Expenditure per patient day equivalent (district hospitals)
- Medical scheme coverage (ave)

Only geographical areas were defined by the names of the provinces and districts.

2. Data Quality Audit & Preprocessing

Following the principles of the World Health Organization' data quality review framework, the data audit was successfully conducted before the examination of the relationship between variables.

The dataset was analysed for any missing values, duplicates, inconsistent data types, and consistency.

No duplicates and missing records were found in the dataset. However we identified a problem with our province columns, it had 43 blank entries. This was because in the original dataset the structure of the table showed us the name of a province once, followed by the districts that belong to that specific province. Since the blanks were not missing values the provincial values were forward-filled.

Descriptive stats was also utilized to examine the numerical variables. The ranges of some

indicators were excessively high. For instance, **% of target tested for TB** ranged from 46% to 390%, while professional nurses per 100000 ranged from approximately 52 to 3834. These unusual values were not removed because we cannot just assume that unusual values are incorrect values. Removing them without enough evidence might result in our outcome being biased or distort the results. Therefore, all 52 districts were included in the analysis.

Additionally, column names were standardised to help improve their usability and consistency . Since the explanatory variables were measured on different scales as percentages, expenditures, and rates per population, they were also standardised for the regression analysis.

3. Exploratory Data Analysis[EDA]

To be able to determine how TB lost to follow up varied across districts and how it correlated to the identified exploratory variables, EDA was utilised. Several EDA were looked at such as correlation matrix, boxplot, descriptive stats, and scatterplots.

The relationships that had the greatest correlation were between TB lost to follow up rate and **% of target tested for TB ($r = -0.521$)** The negative correlation demonstrates that districts having a higher success against their % TB testing target is likely to have a lower loss to follow up rates. Secondly, **medical scheme ($r = 0.4663$)**, which demonstrated a positive correlation, and lastly **TB DS death rate followed with a correlation of ($r = -0.434$)**, which showed a negative relationship.

The rest of the variables had a weaker correlation with lost to follow up rates. **PHC expenditure per PHC headcount had a correlation of -0.228, professional nurses per 100 000 population -0.189, expenditure per patient day equivalent -0.187, TB/HIV co-infected clients on ART rate -0.046, and District Health Services expenditure per capita 0.028.**

These findings indicate that a single health system measure is unlikely to account for districts capacity to retain TB patients.

4. Linear Regression and conclusion:

Utilising 41 of the training data and 11 testing observations, the linear regression model examined the variables together. With the **$R^2 = 0.349$, a mean absolute error = 3.364, and the root mean squared = 3.855**. The model predicted almost 34.9% of the variation in TB lost to follow up rates in the test data. When they were standardised the **Medical scheme had the highest coefficient**, this was followed by % of target tested for TB, and then TB DS death rate. Since the data provided by the predictors may overlap, these coefficients must be assessed with caution.

Overall, the % of target tested for TB, medical aid scheme, and TB DS death rate were strongly related to variation in patient retention. The negative correlation among the TB testing and Lost to follow up rates demonstrate that the district that exceeded their testing target had a higher retention rate. Therefore, these findings identify important variables for

further investigation rather than proving that they directly determine TB patient retention.

REFERENCE

- World Health Organization. *Data Quality Assurance (DQA), Module 2: Desk Review*.
WHO. <https://www.who.int/data/data-collection-tools/health-service-data/data-quality-assurance-dqa/module-2-desk-review>
- Health Systems Trust. District Health Barometer 2024/25. Available at: <https://www.hst.org.za/publications/Pages/HSTDistrictHealthBarometer.aspx> .